

भौतिक अनुसंधान प्रयोगशाला, अहमदाबाद

Physical Research Laboratory Ahmedabad

SUMMER INTERNSHIP PROJECT REPORT

CERTIFICATE

This is to certify that the work embodies in this summer internship project entitled "Atmospheric Visibility Estimation from Ground-Based Imagery" being submitted by **Sakshi Giglani** of **GLS University, Ahmedabad** for partial fulfillment of the requirement for the award of "Certificate" during 19th May 2026 – 19th July 2026. PRL's Summer internship programme 2026 is a record of a bonafide and original piece of work, which was carried out by her under my supervision and guidance in the Space and Atmospheric Sciences Division (SPASC) of PRL, Ahmedabad.

Pand
29th July '26

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PHYSICAL RESEARCH LABORATORY

Project Internship Report

Atmospheric Visibility Estimation from Ground-Based Imagery

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Atmospheric visibility describes the distance over which an object can be distinguished from its background under specified observing conditions. It is an important indicator of the optical state of the lower atmosphere and is influenced by the scattering and absorption of light by aerosols, haze, fog, cloud droplets, dust and other atmospheric constituents. Relative humidity, illumination, sun angle, scene geometry, camera exposure and the contrast of reference objects can also affect how clearly a scene appears.

A reduction in visibility is generally observed as fading of distant structures, weakening of edges, loss of contrast and the appearance of a whitish or grey atmospheric veil. Such changes are relevant to atmospheric research because they provide information about variations in haze, moisture and suspended particles. However, manual visual assessment can be subjective, particularly when the difference between two visibility conditions is small.

Conventional visibility monitoring commonly relies on specialised instruments or trained human observations. Although these methods provide physically meaningful measurements, they may require dedicated equipment, installation and regular maintenance. A ground-based camera offers a complementary research approach because it is comparatively inexpensive, can capture observations repeatedly and preserves the complete scene for later inspection.

When a camera remains fixed at the same location, the viewpoint, background structures and reference landmarks remain approximately unchanged. Variations between images can therefore be studied in terms of atmospheric clarity and illumination. Under clearer conditions, distant objects usually retain stronger contrast and sharper boundaries. Under reduced visibility, the same objects appear faded or less distinguishable from their surroundings.

Artificial intelligence and image-analysis techniques can represent these visual variations numerically and learn their relationship with known visibility values. Modern pretrained image models are particularly useful when the local labelled dataset is small because they can extract general visual information without requiring a large scene-specific training dataset.

The present project, titled **Atmospheric Visibility Estimation from Ground-Based Imagery**, was developed for a fixed observation scene at the Physical Research Laboratory (PRL), Ahmedabad. A GoPro Hero 9 camera was installed at a rooftop observation point facing a consistent reference scene. Images were collected under different atmospheric conditions, together with visibility labels and supporting meteorological information.

The project followed two complementary data stages. A large online image dataset was first used to test whether visibility-related patterns could be learned from outdoor imagery and meteorological information. A smaller local PRL dataset was then used to develop and evaluate a scene-specific visibility-estimation system. The final system accepts a fixed-view ground image and produces an estimated horizontal visibility value in kilometres. The project also investigates the effects of limited data, repeated observations from the same date and an

imbalanced visibility distribution. These aspects are important because the value of the project lies not only in producing a prediction, but also in determining whether the prediction generalises to atmospheric conditions that were not used during training.



Figure 1.1: Fixed-view PRL observation scene used for atmospheric visibility estimation.

1.2 Problem Statement

The practical problem addressed in this project is the estimation of horizontal atmospheric visibility from a single image captured at a fixed PRL observation point. A camera does not measure visibility directly. It records a combination of atmospheric conditions, illumination, camera response and scene characteristics. Changes in haze or moisture may reduce distant contrast, while changes in sunlight or exposure may alter the image even when visibility remains similar. The model must therefore distinguish visibility-related changes from other visual variations.

The local dataset also presents two major challenges. First, the number of independent atmospheric conditions is small because the images were collected across only 23 acquisition dates. Images captured on the same date may share nearly identical lighting and atmospheric conditions. Randomly dividing such images between training and testing could produce overly optimistic performance.

Second, the visibility labels are highly imbalanced. Most local observations were labelled exactly 10 km, while only a small number represented reduced visibility. Consequently, a simple system that always predicts 10 km already performs well on the majority of the dataset. The evaluation must therefore consider both overall prediction error and the ability to respond to reduced-visibility conditions.

The central research question is:

Can the visual appearance of a fixed ground-based PRL scene be used to estimate atmospheric visibility in kilometres for acquisition dates that were not used during model training?

The project further examines whether image features, meteorological variables or their combination provide the most reliable local predictions, and whether information obtained from a larger online dataset can support the smaller PRL dataset.

1.3 Objectives

The main objective of the project is to develop and evaluate a reproducible research system for estimating atmospheric visibility from fixed-view ground-based images.

The specific objectives are:

1. To collect and organise a labelled image dataset from a fixed PRL observation point.
2. To align each image with visibility, date, time, location and available meteorological information.
3. To prepare a larger online source dataset for initial visibility-model development.
4. To extract informative visual representations from outdoor images using a pretrained image encoder.
5. To compare weather-only, image-only and image–weather fusion approaches.
6. To develop a regression model suitable for the small local PRL dataset.
7. To evaluate model performance on complete acquisition dates that were not used during training.
8. To compare the learned models with a meaningful constant visibility baseline.
9. To evaluate both overall prediction performance and reduced-visibility behaviour.
10. To create a reproducible workflow for data verification, model training, evaluation and single-image inference.

1.4 Scope and Contributions

The completed project is a research prototype for atmospheric visibility estimation from one fixed scene at PRL, Ahmedabad. It is designed for daytime images captured using the same camera, viewpoint and reference scene. The visibility range represented in the local dataset is 4 km to 10 km.

The project does not claim to provide a universal visibility model for all locations, cameras or weather conditions. It has not been validated for night-time imagery, moving cameras, changing scenes, rain on the lens or severe occlusion. It is also not intended to replace a calibrated visibility instrument or to support aviation, road-safety or other safety-critical decisions.

Meteorological variables were collected and evaluated because humidity, temperature, solar conditions and time may influence atmospheric appearance. However, the final local model uses image information because the available weather variables did not provide a stable improvement on unseen acquisition dates. The weather records remain valuable for analysis and for future evaluation using a larger local dataset.

The principal contributions of the project are:

- A structured local dataset of 127 fixed-view PRL images collected across 23 acquisition dates.
- A verified 14,906-image source dataset used to establish a large-scale visibility-learning proof of concept.

- A complete image-to-visibility workflow based on pretrained visual features, dimensionality reduction and regression.
- A comparison of weather-only, image-only, fusion and source-transfer approaches.
- An evaluation design that keeps complete acquisition dates separate during training and testing.
- A comparison with a strong constant 10 km reference model.
- A performance analysis that distinguishes overall error from reduced-visibility behaviour.
- A reusable software workflow for data preparation, model evaluation and visibility prediction.

The project therefore provides both a working visibility-estimation prototype and a documented baseline for future data collection and model improvement at PRL.

1.5 Organization of the Report

The remainder of the report is organised into seven chapters.

Chapter 2 describes the source and local datasets, the PRL camera setup, recorded parameters, data preparation and visibility distribution.

Chapter 3 explains the proposed visibility-estimation methodology, including image-feature extraction, dimensionality reduction, regression models and the final selected pipeline.

Chapter 4 presents the model-development process and the evaluation methodology used to test performance on unseen acquisition dates.

Chapter 5 reports the source-domain and PRL results and discusses the effects of dataset size, label imbalance and reduced-visibility performance.

Chapter 6 documents the software implementation, major scripts, model outputs, prediction workflow and reproducibility checks.

Chapter 7 presents the limitations of the completed work and the main directions for future data collection, validation and system development.

Chapter 8 summarises the project, its main findings and the final conclusions.

CHAPTER 2

DATA COLLECTION AND DATASET PREPARATION

2.1 Data Sources

The project used two primary datasets: a large online source dataset and a smaller local PRL target dataset.

The online dataset was used to establish whether visibility-related information could be learned from outdoor images and meteorological variables when a comparatively large number of labelled observations was available. It also supported the development of the image-processing and feature-extraction workflow.

The local PRL dataset represented the actual target environment of the project. It contained fixed-view images captured from one rooftop observation point at the PRL campus in Ahmedabad. This dataset was used for local model development, model selection and final evaluation.

The two datasets served different purposes and were not treated as directly interchangeable. The online images were captured using different cameras, scenes, geographic locations and visibility distributions. The PRL images were collected from one consistent scene and therefore provided a more controlled but much smaller target dataset.

Table 2.1: Overview of the datasets used in the project

Characteristic	Online source dataset	PRL target dataset
Dataset source	SkyFinder outdoor webcam data	Locally collected PRL images
Number of images	14,906	127
Number of observation locations	5 selected source sites	1 fixed PRL viewpoint
Number of acquisition dates	860	23
Main purpose	Source-domain model development	Local training and evaluation
Image type	Outdoor webcam scenes	Fixed-view GoPro images
Visibility information	Source visibility labels	Local exact visibility labels
Weather information	Available weather and NASA POWER solar data	Temperature, humidity and solar intensity

2.2 SkyFinder Source Dataset

The source-domain stage used imagery derived from the SkyFinder outdoor webcam dataset. A cleaned working subset of 14,906 images was created from five coordinate-defined sites. The selected observations covered 860 unique dates between 15 June 2012 and 25 October 2014.

The source records contained image identifiers, visibility values, temperature, humidity, location, date and time. Solar information was added using data obtained from NASA POWER.

The solar records were retrieved by site and year and were matched to the image records using location and date.

Table 2.2: Site-wise distribution of the SkyFinder working subset

Latitude	Longitude	Number of images
29.272958	-94.850840	3,687
33.749745	-84.388369	3,491
41.979700	-88.406100	3,241
32.981487	-111.529717	3,232
46.383333	11.850000	1,255
Total		14,906

The source dataset contained 51 distinct visibility values. However, it was also imbalanced: 11,447 images, representing approximately 76.8% of the source subset, had a visibility value of 16.1 km. The source dataset was therefore useful for testing visibility-learning approaches, but its label distribution was not fully balanced.

After cleaning and integration, the source master dataset contained 14,906 valid rows. Verification confirmed that there were no duplicate image identifiers, no missing selected image files and no missing merged solar values.

NASA POWER solar data were included to provide additional environmental context. During later source-to-target comparison, a substantial difference was observed between the numerical scale of the source solar field and the local PRL solar-intensity measurements. The median source solar value was approximately 4.8029, while the median PRL solar intensity was 798 W m^{-2} . The ratio between the medians was approximately 166.

This difference indicated that the two solar variables likely represented different units, aggregation periods or measurement conventions. Source and PRL solar values were therefore not assumed to be directly interchangeable. The difference was documented so that source-domain results and local fusion experiments could be interpreted cautiously.

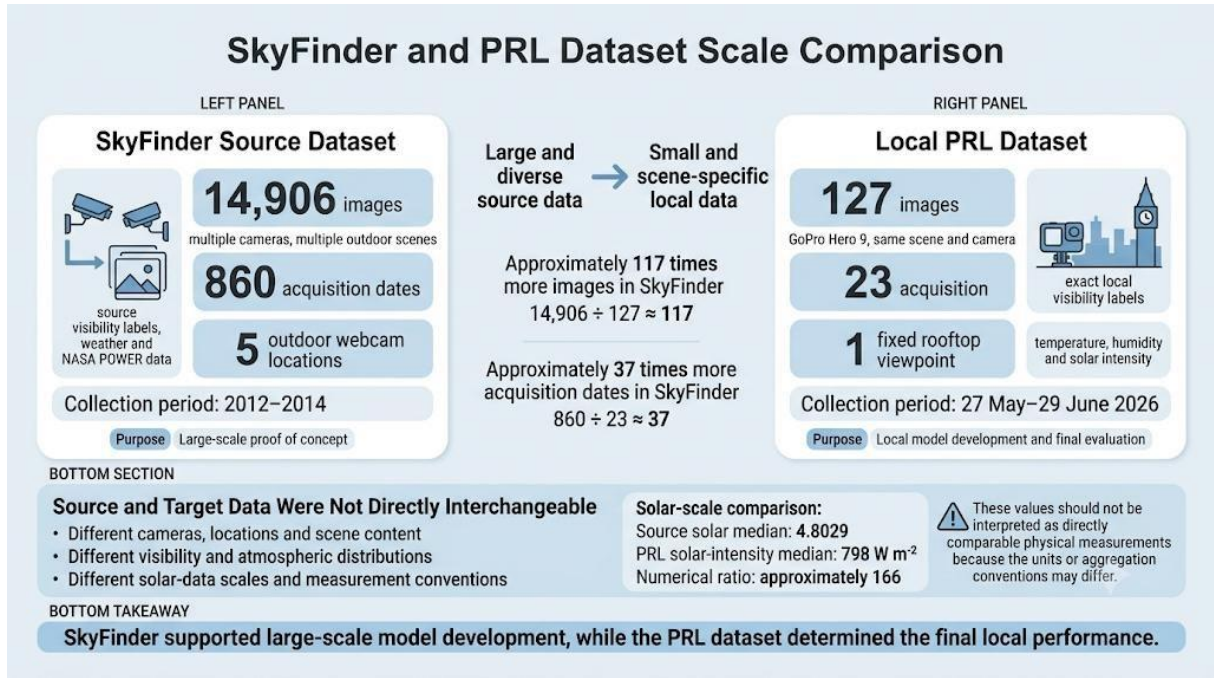


Figure 2.1: Scale comparison between the SkyFinder source dataset and the local PRL dataset.

The source-domain stage was treated as a proof of concept rather than as the final estimate of PRL performance. The final scientific conclusions were based on the local PRL evaluation.

2.3 PRL Local Data Collection

The local dataset was collected from a fixed rooftop observation point on the PRL campus in Ahmedabad. A GoPro Hero 9 camera was used to capture the same outdoor scene throughout the collection period. The fixed setup ensured that the camera position, viewing direction and reference structures remained consistent. The PRL clock tower and the surrounding distant skyline acted as visual reference features. Changes in their contrast and clarity could therefore be examined under different atmospheric conditions.

The approximate observation coordinates were:

- **Latitude:** 23.036° N
- **Longitude:** 72.544° E

The metadata also contained a reference distance of approximately 265 m to the clock tower landmark. Images were collected across 23 acquisition dates between 27 May and 29 June 2026. Observations were made during afternoon hours, covering approximately 2:00 PM to 5:59 PM. The intended observation slots were 2 PM, 3 PM, 4 PM and 5 PM.

Table 2.3: PRL data-collection configuration

Item	Configuration
Observation site	Physical Research Laboratory, Ahmedabad
Camera	GoPro Hero 9
Camera placement	Fixed rooftop setup
Observation scene	PRL clock tower and surrounding skyline

Approximate latitude	23.036° N
Approximate longitude	72.544° E
Reference distance	265 m
Collection period	27 May–29 June 2026
Observation time	Approximately 2 PM–5:59 PM
Number of images	127
Acquisition dates	23



Figure 2.2/2.3: GoPro Hero 9 rooftop setup used for PRL image collection.

The use of one fixed viewpoint reduced variations caused by changes in camera location and scene geometry. However, image appearance could still be affected by sunlight, clouds, exposure and local environmental conditions. These factors were considered during model development and evaluation.

2.4 Recorded Image and Atmospheric Parameters

Each local image was associated with a structured metadata record. The recorded information was used to align the image with its visibility label and observation conditions.

Table 2.4: Parameters recorded for each PRL observation

Parameter	Description	Role in the project
Image name	Unique identifier for the image	Alignment of image, metadata and features
Date	Date of acquisition	Evaluation grouping variable
Time	Local image-capture time	Observation reference
Hour	Numerical hour of acquisition	Candidate model input
Visibility	Horizontal visibility in kilometres	Prediction target
Temperature	Ambient temperature in °C	Meteorological input
Relative humidity	Relative humidity in percentage	Meteorological input
Solar intensity	Local solar intensity in W m^{-2}	Illumination-related input
Latitude	Observation latitude	Location record
Longitude	Observation longitude	Location record

Distance	Distance to reference landmark	Scene documentation
----------	--------------------------------	---------------------

The numerical ranges observed in the final local dataset are summarised below.

Table 2.5: Statistical summary of the PRL target dataset

Variable	Minimum / count	Median	Maximum
Images	127	–	–
Acquisition dates	23	–	–
Temperature (°C)	30.0	32.9	40.0
Relative humidity (%)	57	73	89
Solar intensity (W m^{-2})	317	798	1,194
Visibility (km)	4	10	10
Hour of day	14	15	17

The meteorological values were retained because atmospheric appearance can be affected by temperature, humidity and illumination. Weather-only and image–weather fusion approaches were later evaluated. Their inclusion in the dataset also preserves the possibility of future analysis using a larger and more diverse local collection.

2.5 Data Cleaning, Alignment and Verification

A consistent image and metadata structure was created before model development. The local images were renamed using sequential identifiers from PRL_0001.jpg to PRL_0127.jpg. Stable identifiers allowed every image to be matched reliably with its metadata, visibility label and extracted numerical features.

The preparation process included the following steps:

1. Standardisation of image names and file formats.
2. Combination of image records, visibility labels and meteorological observations.
3. Verification that every metadata row referred to an available image.
4. Checking for duplicate image identifiers.
5. Checking numerical fields for missing or invalid values.
6. Sorting and aligning image names before creating model-ready arrays.
7. Creating a date-group variable so that all observations from the same acquisition date remained together during evaluation.
8. Preserving raw, intermediate and final data files separately.

The final verification confirmed that the local dataset contained 127 metadata rows, 127 image records and 127 aligned feature rows. No duplicate image identifiers or missing numerical values were found.

Table 2.6: Final PRL dataset verification results

Verification check	Observed result	Status
Metadata rows	127	Pass
Available image records	127	Pass
Aligned feature rows	127	Pass
Acquisition dates	23	Pass
Duplicate image identifiers	0	Pass

Missing numerical values	0	Pass
Valid visibility labels	4, 5, 9 and 10 km	Pass
Dominant-label share	82.7% at 10 km	Interpretation warning
Source–target solar median ratio	Approximately 166	Scaling warning

The verification process was necessary because a mismatch between image order, labels and feature rows would invalidate model training. The final aligned dataset was therefore checked before every major modeling stage.

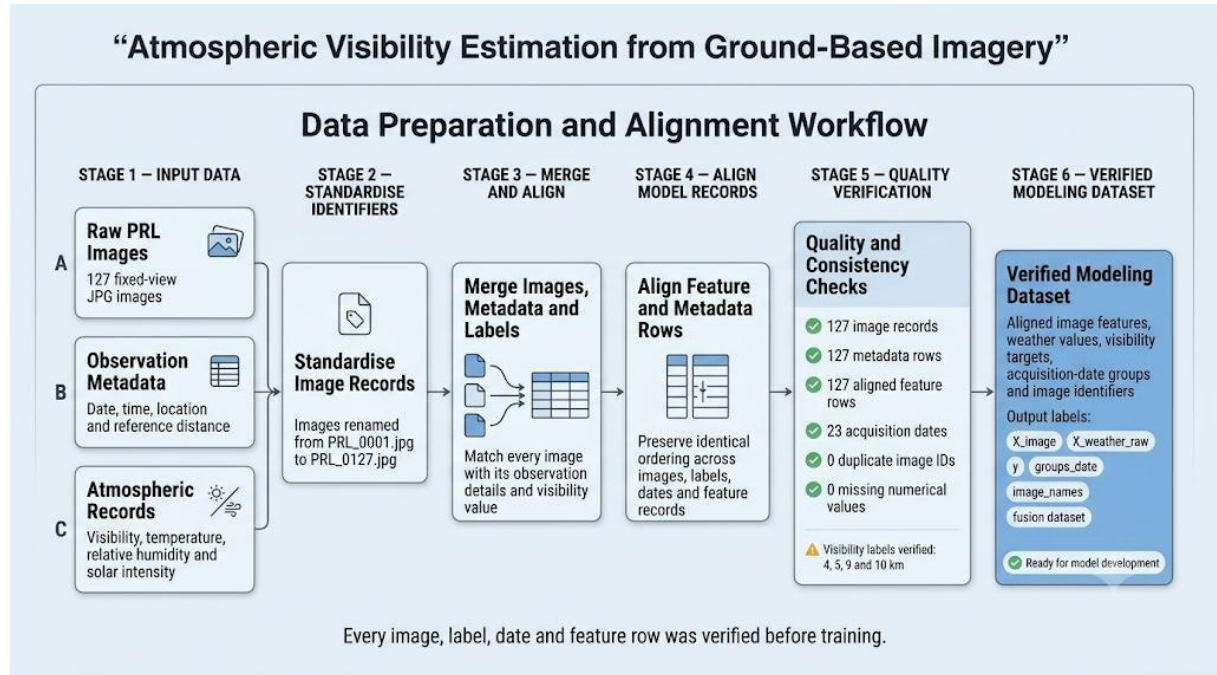


Figure 2.4: Data preparation and alignment workflow from raw images and metadata to the verified modeling dataset.

2.6 Visibility Distribution and Dataset Imbalance

The final PRL dataset contained four exact visibility values: 4 km, 5 km, 9 km and 10 km.

Table 2.7: Exact PRL visibility-label distribution

Visibility label	Number of images	Percentage
4 km	1	0.8%
5 km	10	7.9%
9 km	11	8.7%
10 km	105	82.7%
Total	127	100.0%

The distribution is highly imbalanced. Only 22 observations had visibility below 10 km, while 105 observations were labelled exactly 10 km. The 4 km condition was represented by only one image.

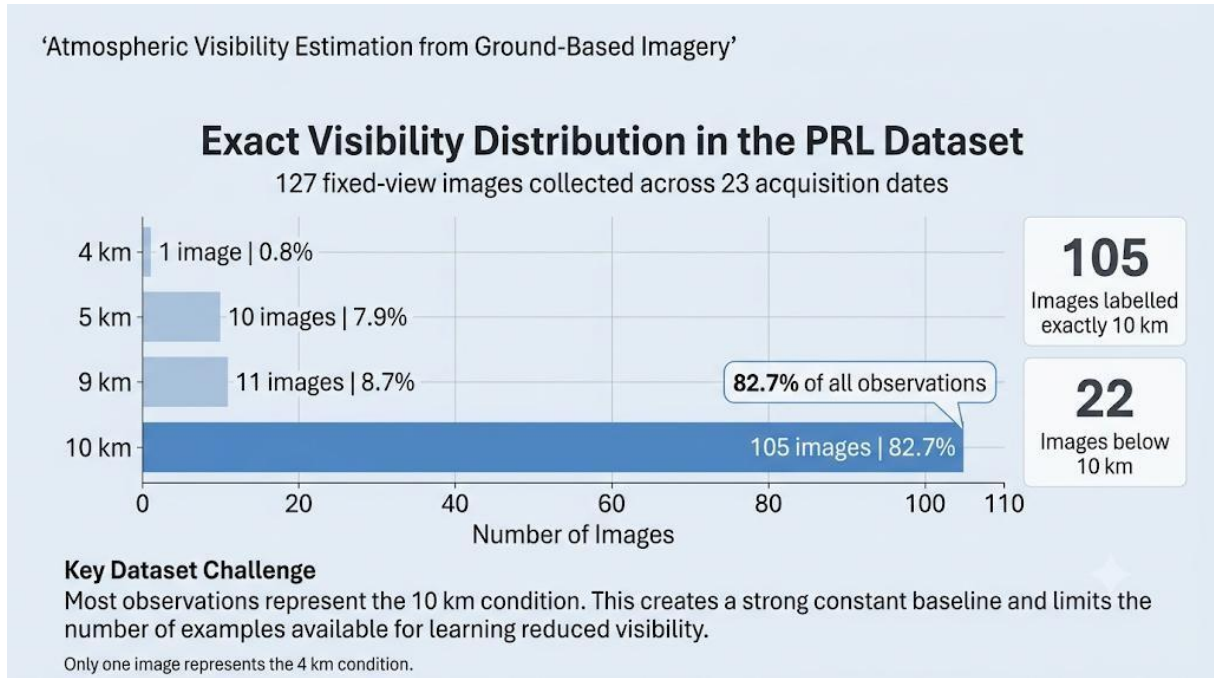


Figure 2.5: Distribution of visibility labels in the local PRL dataset.

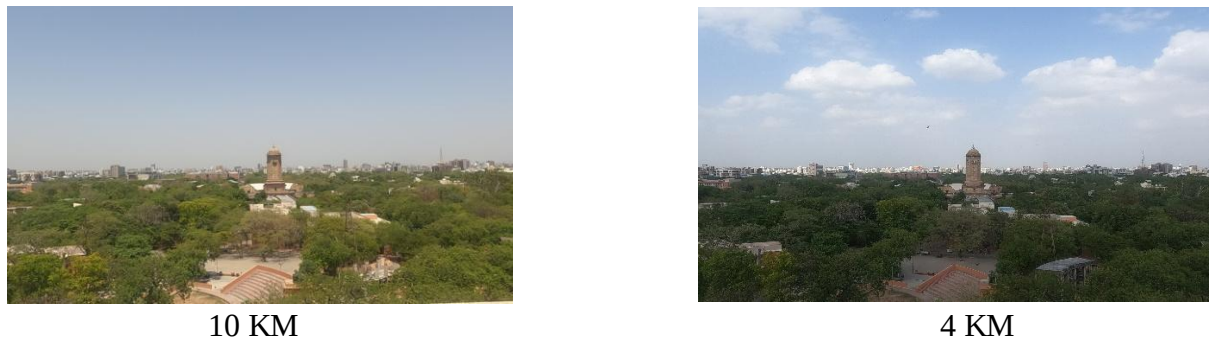


Figure 2.6: Visual comparison between the 4 km and 10 km PRL observations.

Because the camera and scene remained fixed, differences between visibility conditions were primarily visible in distant contrast, edge clarity and the fading of skyline structures. In some cases, these differences were subtle. The model therefore had to detect relatively small visual changes from a limited number of reduced-visibility examples.

The imbalance also created a strong constant baseline. A model that always predicts 10 km produces zero error for 105 of the 127 images. Its errors are restricted to the one 4 km image, ten 5 km images and eleven 9 km images. This explains why the constant reference performs strongly on overall average error.

The imbalance does not imply that the images contain no visibility information. A learned model may improve predictions for reduced-visibility observations while making small errors on some of the many 10 km observations. For this reason, later evaluation considers both overall model performance and performance specifically for observations below 10 km.

The effective amount of independent information is also smaller than the total image count. The 127 images were collected across 23 dates, and images from the same date share related

atmospheric and illumination conditions. Acquisition date was therefore retained as an explicit grouping variable for the final evaluation. Overall, the local dataset provides a carefully aligned fixed-scene research baseline. Its main strengths are the consistent observation geometry, exact image-to-metadata alignment and availability of both image and meteorological information. Its main limitations are the small number of independent dates and the concentration of observations at 10 km. These limitations directly influenced the modeling and evaluation decisions described in the following chapters.

CHAPTER 3

VISIBILITY ESTIMATION METHODOLOGY

3.1 Overall Workflow

The proposed system estimates horizontal atmospheric visibility from a fixed-view ground-based image of the PRL scene. The final operational workflow consists of four principal stages: **Input image** → **DINOv2 feature extraction** → **Principal Component Analysis** → **Support Vector Regression** → **Estimated visibility**

The input image is first transformed into a numerical visual representation using a pretrained DINOv2 image encoder. Since the resulting representation contains 768 features, Principal Component Analysis is used to reduce its dimensionality. The reduced feature vector is then provided to a Support Vector Regression model, which produces the final visibility estimate in kilometres.

During development, the project evaluated three broad feature configurations:

1. Meteorological variables only
2. Image features only
3. Combined image and meteorological features

The final PRL model uses image features only. The meteorological variables were retained in the dataset and evaluated experimentally, but their inclusion did not provide a consistent improvement when the system was tested on unseen acquisition dates.

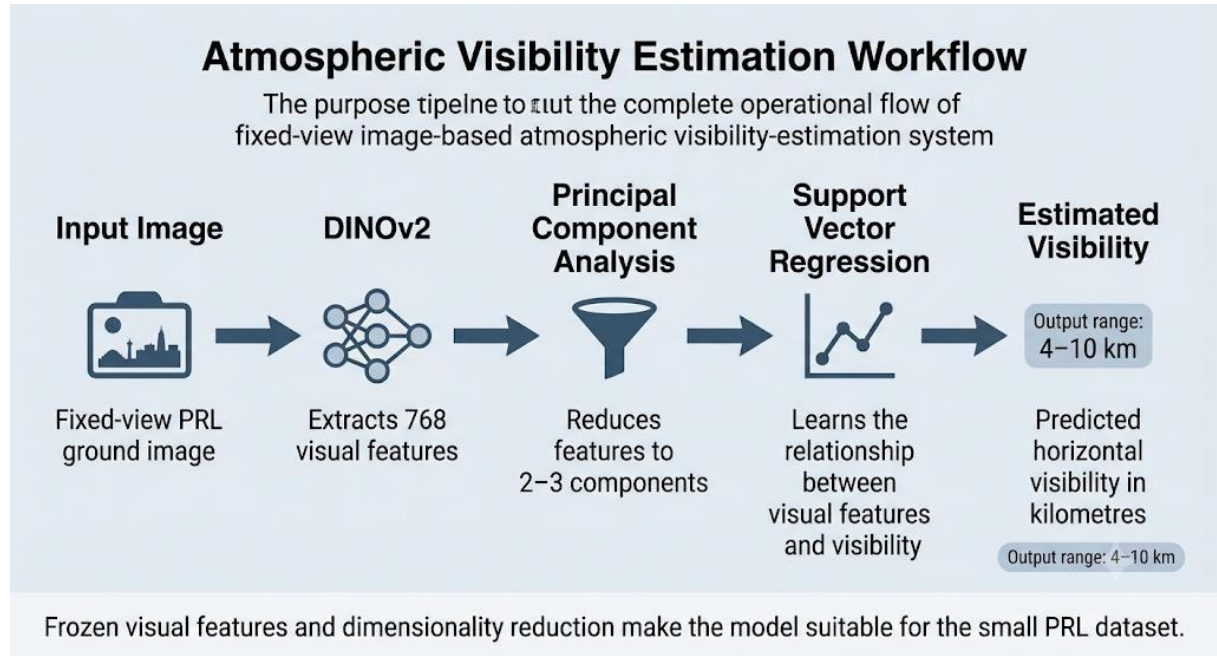


Figure 3.1: End-to-end atmospheric visibility-estimation workflow.

3.2 DINOv2 Feature Extraction

DINOv2 is a self-supervised visual foundation model designed to produce general-purpose image representations. The project used the pretrained **DINOv2 ViT-B/14** model as a frozen feature encoder.

Each image was resized and normalised using the preprocessing expected by the encoder. The processed image was then passed through the network to generate a **768-dimensional embedding**. This numerical vector represents visual characteristics of the complete image, including variations associated with contrast, distant-scene clarity, illumination and haze-related appearance.

The DINOv2 model was not fine-tuned on the local PRL images. Fine-tuning a large vision transformer using only 127 images and 23 independent acquisition dates would introduce a substantial risk of overfitting. Using the encoder in frozen form allowed the project to benefit from a pretrained visual representation while keeping the number of locally trained parameters small. The same DINOv2 model and preprocessing route were used for the source and PRL image features. This ensured that the embeddings were produced consistently.

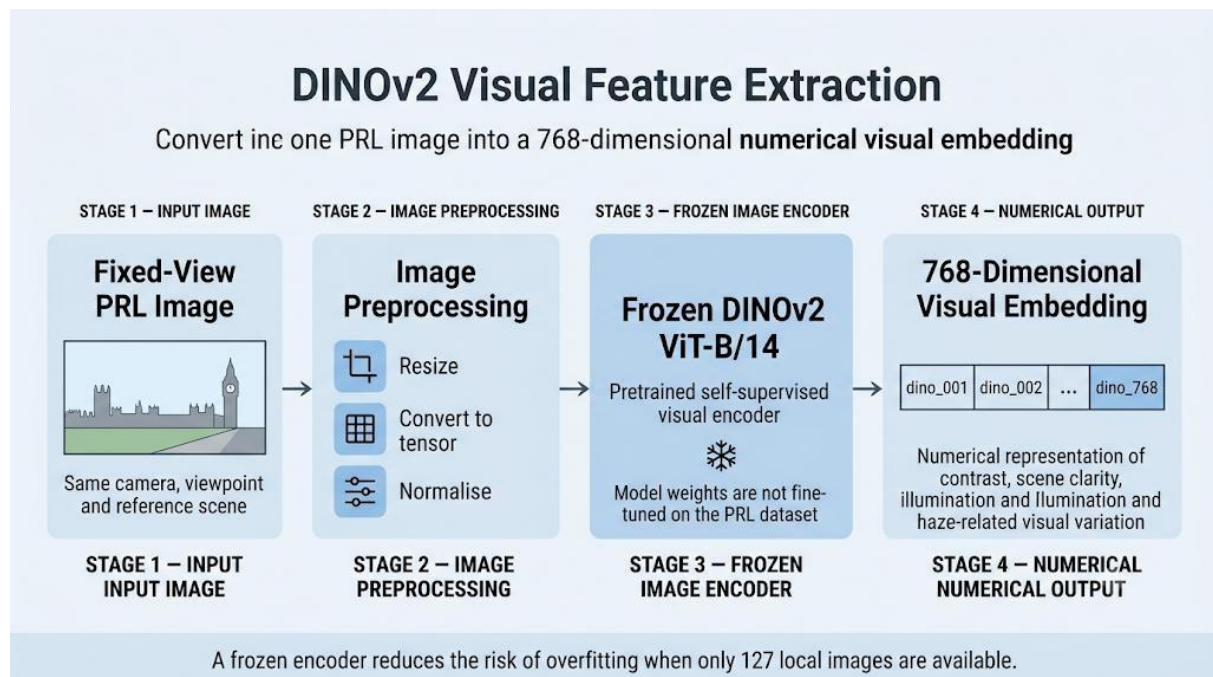


Figure 3.2: Extraction of a 768-dimensional visual embedding using frozen DINOv2 ViT-B/14.

3.3 Principal Component Analysis

The 768-dimensional DINOv2 representation is large compared with the number of available PRL observations. Training a regression model directly on all embedding dimensions could allow the model to fit image-specific variations that do not generalise to new dates.

Principal Component Analysis was therefore used to transform the original embedding into a lower-dimensional representation. PCA identifies directions of variation in the training features and expresses each image using a smaller number of uncorrelated components.

The main benefits of PCA in this project were:

- reducing the number of input variables;
- limiting overfitting;
- removing redundant feature directions;
- reducing computational complexity; and
- making regression more suitable for the small local dataset.

Early experiments evaluated several PCA dimensions. The focused model search showed that the strongest final candidates generally used only **two or three principal components**. This indicates that the most useful visibility-related variation in the fixed PRL scene was concentrated in a small number of visual directions.

PCA was fitted only on the training portion of each validation split. The test images were transformed using the PCA model learned from the corresponding training data. This prevented information from the held-out data from influencing feature reduction.

3.4 Feature Configurations and Regression Models

Table 3.1: Feature configurations evaluated

Configuration	Input features	Purpose
Weather-only	Temperature, relative humidity, solar intensity and hour	Evaluate whether recorded atmospheric variables predict visibility without image information
Image-only	DINOv2 embedding followed by PCA	Estimate visibility from visual scene appearance
Image-weather fusion	PCA image features combined with scaled meteorological features	Test whether image and weather information provide complementary local information

The meteorological inputs included temperature, relative humidity, solar intensity and local hour. Additional deterministic weather features, such as cyclic time representation and humidity-related quantities, were also considered during focused development.

The following regression families were evaluated:

- **Ridge Regression**, as a regularised linear reference;
- **XGBoost**, for nonlinear tree-based prediction;
- **Support Vector Regression**, for nonlinear regression with controlled model complexity; and
- **constant 10 km prediction**, as the main reference baseline.

Source-transfer experiments were also performed by combining information from the source and target datasets using a weighted Ridge model. This approach did not improve the final PRL generalisation and was retained only as a comparison model.

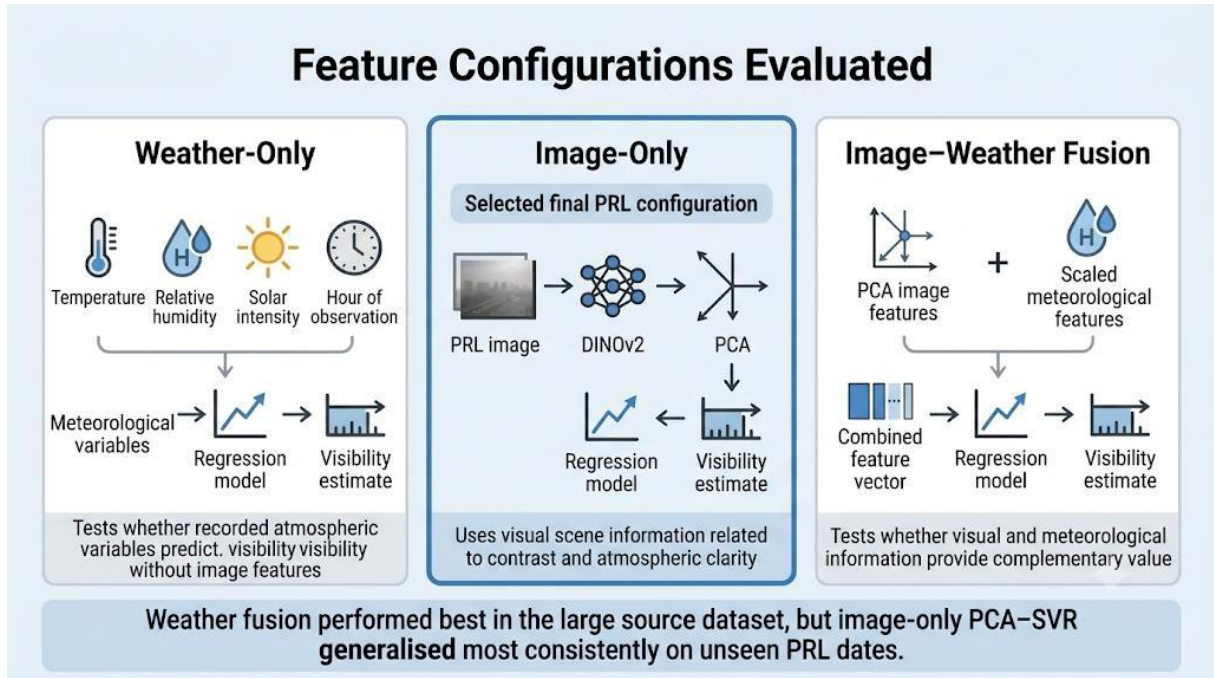


Figure 3.3: Weather-only, image-only and image-weather fusion configurations.

3.5 Final PCA-SVR Pipeline

The final target-domain model family was **image-only PCA-SVR**. The selected architecture consists of the following processing sequence:

1. Load the new PRL image.
2. Apply the configured resize and normalisation procedure.
3. Extract a 768-dimensional DINOv2 embedding.
4. Apply the image-feature scaler stored in the model bundle.
5. Transform the scaled embedding using the stored PCA model.
6. Pass the reduced features to the trained SVR model.
7. Clip the prediction to the observed training range of **4–10 km**.
8. Return the estimated visibility and model information.

Support Vector Regression was selected because it can model nonlinear relationships while remaining appropriate for a small dataset. The final SVR candidates used a radial basis function kernel. Their principal hyperparameters were:

- **C**, which controls the penalty for prediction errors;
- **epsilon**, which defines the width of the error-insensitive region; and
- the number of retained PCA components.

Table 3.2: Final system configuration

Component	Final configuration
Image encoder	Frozen DINOv2 ViT-B/14
Embedding dimension	768
Feature scaling	Standardisation fitted on training data
Dimensionality reduction	PCA
Final PCA range	2–3 components
Regressor	RBF Support Vector Regression
Model input	Image features

Output	Predicted horizontal visibility in kilometres
Prediction range	4–10 km

The final architecture was selected through the grouped evaluation procedure described in Chapter 4. The architecture should therefore be interpreted as the model family that generalised most consistently within the available PRL data, rather than as a universal visibility model.

CHAPTER 4

MODEL DEVELOPMENT AND EVALUATION

METHODOLOGY

4.1 Source-Domain Proof of Concept

The first modeling stage used the 14,906-image SkyFinder working subset. Its purpose was to determine whether outdoor image features and meteorological information contained learnable relationships with visibility when a substantially larger dataset was available.

Three XGBoost baselines were developed:

- weather-only;
- DINOv2 image-only; and
- image-weather fusion.

The source experiment established the image-processing, feature-extraction and fusion workflow. It also demonstrated the value of combining visual and meteorological information within the source domain. The corresponding performance values are presented in Chapter 5. The source-domain results were not treated as direct estimates of PRL performance. The source and PRL datasets differed in scene content, cameras, locations, visibility ranges, weather distributions and solar-data conventions. The source stage was therefore considered a **proof of concept**, while final claims were based only on the PRL held-out-date evaluation.

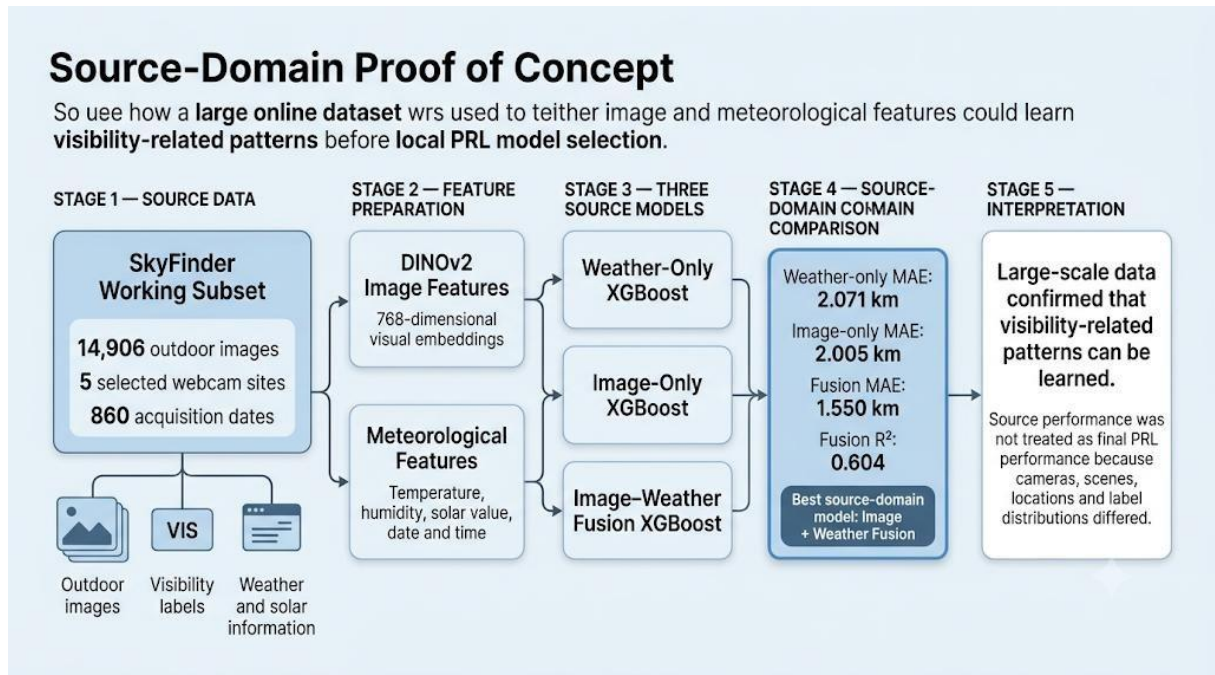


Figure 4.1: Source-domain proof-of-concept workflow using weather, image and fusion models.

4.2 PRL Model Development

Initial PRL experiments compared weather-only, raw-image, fusion and PCA-based models. These early experiments used shuffled cross-validation and were useful for identifying general trends, but they were not used as final performance estimates. The early results showed that direct use of all 768 DINOv2 dimensions produced unstable local models. PCA improved the behaviour of both image-only and fusion approaches by reducing model complexity.

The development workflow was then redesigned to keep complete acquisition dates together. Ridge, XGBoost, image-only SVR and fusion SVR models were compared using grouped validation. A constant 10 km predictor was included because 105 of the 127 local observations had that exact label.

A focused development search evaluated **352 candidate configurations**, including:

- PCA dimensions of 1, 2, 3, 5, 8, 10 and 15;
- RBF-SVR C values of 0.3, 1, 3, 10 and 30;
- epsilon values of 0.25, 0.50, 0.75 and 1.00;
- image-only and image-weather fusion modes; and
- weighted Ridge alternatives.

Predictions were restricted to the 4–10 km target range.

This broad search identified PCA–SVR as the most suitable learned-model family. However, the same grouped predictions had been used to compare all 352 candidates. Reporting the best candidate from that search as the final result would underestimate model-selection uncertainty. A smaller set of **20 candidates** was therefore frozen before the final nested evaluation.

Table 4.1: Frozen final PCA–SVR candidate grid

Model family	PCA components	C values	Epsilon values	Candidates
Image SVR	2 or 3	10 or 30	0.25, 0.50 or 0.75	12
Fusion SVR	2 or 3	3 or 10	0.25 or 0.50	8
Total				20

4.3 Acquisition-Date-Wise Nested Evaluation

Images captured on the same date often shared similar lighting and atmospheric conditions. A random image split could place nearly identical conditions in both the training and test sets. To avoid this, the final evaluation used acquisition date as the grouping variable. The outer evaluation used **Leave-One-Group-Out**, where one complete acquisition date was held out as the test set and the remaining dates were used for training. This was repeated until every one of the 23 dates had served once as the unseen test group.

Within each outer-training set, a five-fold **GroupKFold** procedure selected the model configuration. Complete dates were also kept together in the inner folds.

Two separate selection objectives were used:

1. minimise overall MAE; and
2. minimise balanced-regime MAE.

For every training split, the following operations were performed using training data only:

- image-feature standardisation;
- PCA fitting;

- weather-feature processing and scaling for fusion models;
- SVR fitting; and
- hyperparameter selection.

The selected model was then applied to the untouched outer test date. The 23 outer-fold predictions were combined to create one out-of-fold prediction for every PRL image. These predictions form the authoritative final evaluation results. Models trained later on all 127 observations were created only for future inference. Their training performance was not reported as test performance.

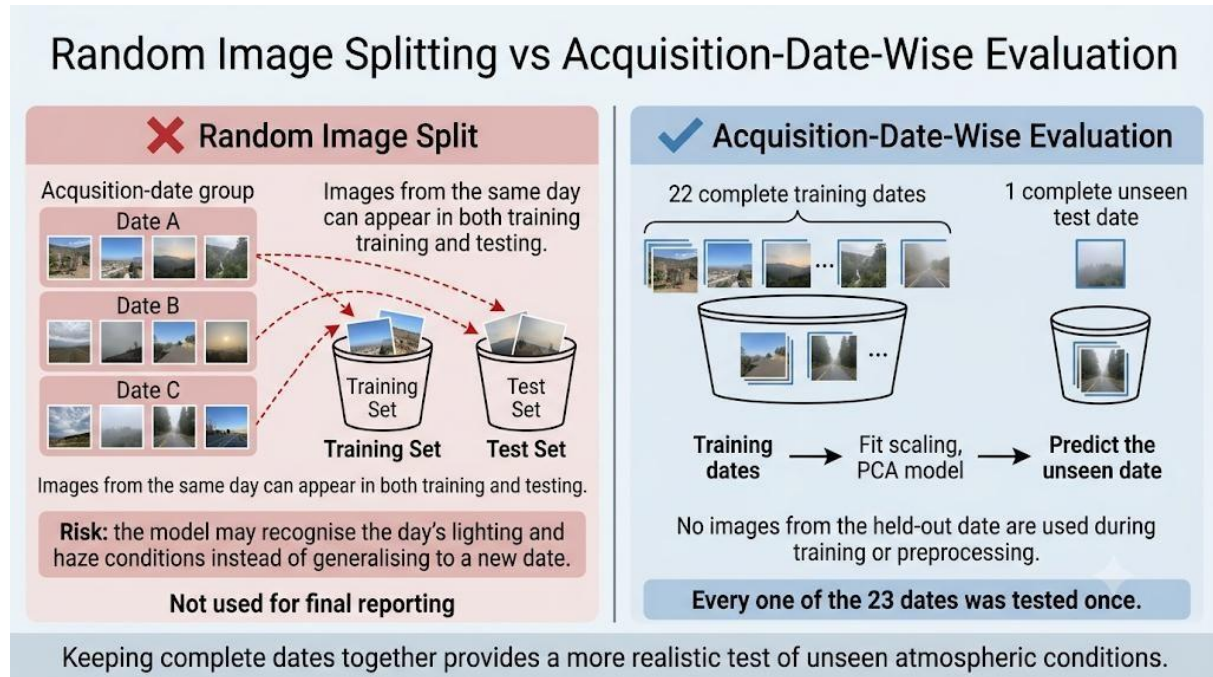


Figure 4.2: Random image splitting compared with acquisition-date-wise evaluation.

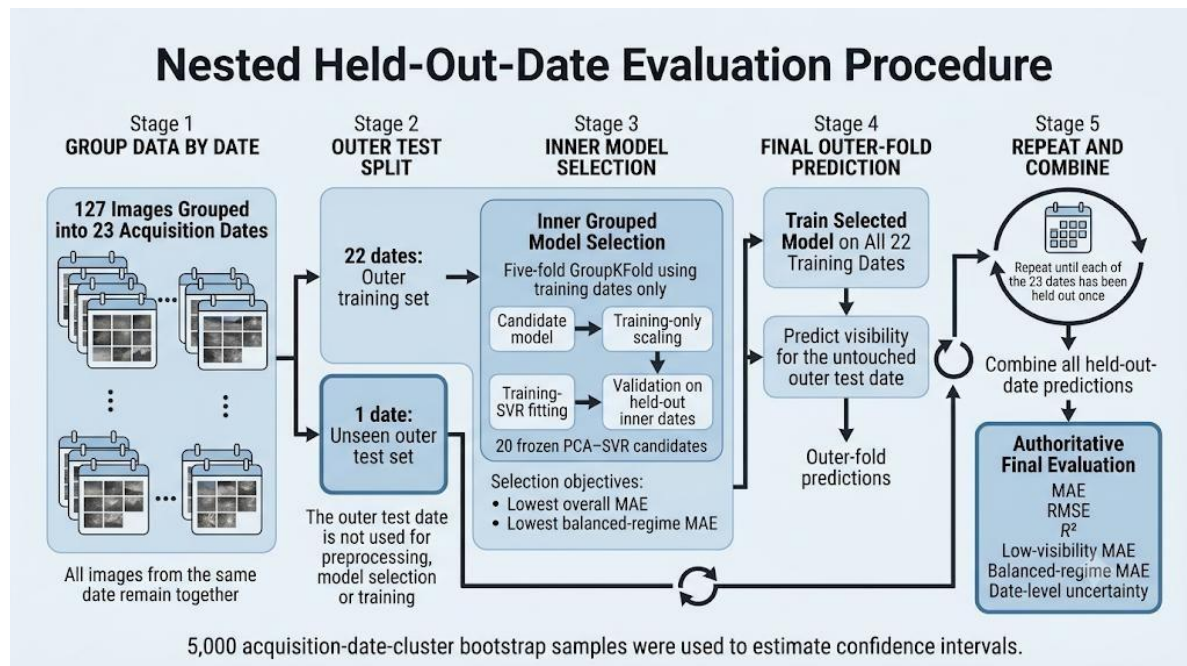


Figure 4.3: Nested held-out-date evaluation procedure.

4.4 Evaluation Metrics

No single metric fully describes performance on the imbalanced PRL dataset. The final evaluation therefore used complementary overall, regime-specific and date-level measures. MAE was treated as the primary sample-weighted measure because it directly expresses the average error in kilometres. RMSE was included to emphasise larger errors, while R^2 was used as a variance-explanation diagnostic. Low-visibility and balanced-regime MAE were essential because overall MAE strongly favoured the dominant 10 km label.

Table 4.2: Evaluation metrics and interpretation

Metric	Meaning	Desired direction
MAE	Average absolute prediction error across all images	Lower
RMSE	Error measure that gives more weight to large deviations	Lower
R^2	Proportion of target variation explained by the predictions	Higher
Low-visibility MAE	MAE for observations below 10 km	Lower
10 km-label MAE	MAE for observations labelled exactly 10 km	Lower
Balanced-regime MAE	Equal average of low-visibility MAE and 10 km-label MAE	Lower
Date-macro MAE	Average of the MAE calculated separately for each acquisition date	Lower
Mean bias	Average signed prediction error	Closer to zero

4.5 Bootstrap Uncertainty and Paired Comparison

The 127 images were not treated as 127 fully independent atmospheric observations. Images from the same date were related and were therefore resampled together during uncertainty estimation. The final analysis used **5,000 acquisition-date-cluster bootstrap iterations**. In each iteration, the 23 acquisition dates were sampled with replacement, and all images belonging to each selected date were included.

This procedure produced confidence intervals that reflect date-to-date variability while preserving the dependence among images collected on the same date.

Model comparisons against the constant 10 km reference were paired. The same sampled dates were used for the learned model and the reference model in every bootstrap iteration.

The paired difference was defined as:

$$\text{Difference} = \text{Learned Model Error} - \text{Constant Baseline Error}$$

For error metrics:

- a negative difference favoured the learned model;
- a positive difference favoured the constant baseline; and
- a confidence interval excluding zero indicated a supported difference between the two procedures.

This evaluation design separated model development from final testing, prevented same-date overlap, limited preprocessing leakage and provided date-level uncertainty estimates. The resulting performance values and their interpretation are presented in Chapter 5.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Source-Domain Results

The source-domain experiment used the 14,906-image SkyFinder working subset to determine whether visibility-related patterns could be learned from outdoor images and meteorological information. Weather-only, DINOv2 image-only and image–weather fusion models were evaluated using XGBoost.

The image-only model performed better than the weather-only model, indicating that visual scene information contained useful visibility-related patterns. The fusion model achieved the best source-domain performance across all three metrics. This result showed that image and meteorological information were complementary within the large source dataset.

Table 5.1: Source-domain model performance

Model	MAE (km)	RMSE (km)	R ²
Weather-only	2.071	4.928	0.429
DINOv2 image-only	2.005	4.776	0.463
Image–weather fusion	1.550	4.102	0.604

However, these values were not treated as estimates of PRL performance. The source and PRL datasets differed in camera type, scene content, geographic location, visibility distribution and environmental measurements. The source experiment therefore served as a proof of concept rather than direct validation of the final local system.

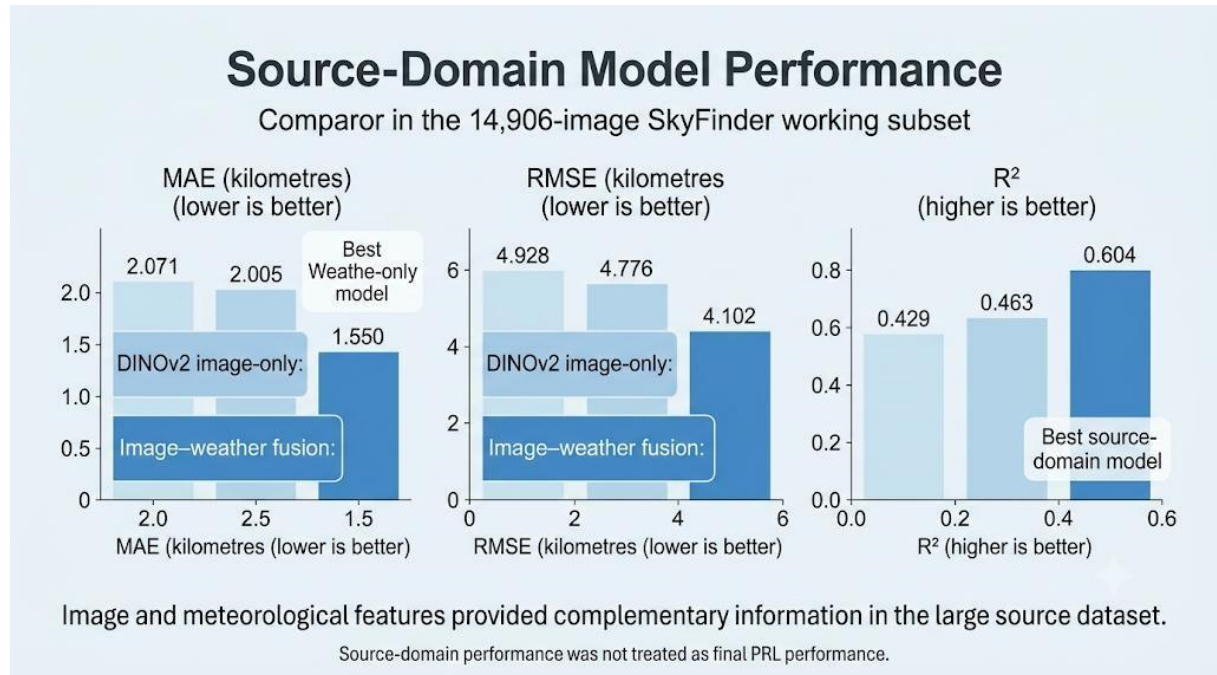


Figure 5.1: Source-domain comparison of weather-only, image-only and image–weather fusion models.

5.2 Final PRL Results

Final PRL performance was calculated from predictions made on complete acquisition dates excluded from model training and model selection. The main comparison included the constant 10 km baseline, two nested PCA–SVR procedures and the principal fixed comparison models.

Table 5.2: Final nested PRL model comparison

Model	MAE (km)	RMSE (km)	R ²	Low-visibility MAE (km)	10 km MAE (km)	Balanced MAE (km)
Constant 10 km baseline	0.528	1.529	−0.135	3.045	0.000	1.523
MAE-selected PCA–SVR	0.651	1.454	−0.026	2.754	0.211	1.482
Balanced PCA–SVR	0.731	1.422	0.019	2.491	0.362	1.426
Fusion PCA–XGBoost	0.893	1.487	−0.073	2.510	0.554	1.532
Image PCA–Ridge	0.949	1.530	−0.137	2.480	0.628	1.554
Weather Ridge	0.958	1.519	−0.120	2.609	0.612	1.611
Transfer-weighted Ridge	1.037	1.627	−0.286	2.644	0.700	1.672

The constant baseline achieved the lowest overall MAE of 0.528 km. The nested PCA–SVR selected using overall MAE achieved 0.651 km, while the balanced PCA–SVR achieved 0.731 km. The balanced model nevertheless obtained the lowest RMSE, 1.422 km, and the only positive R², 0.019, although the value remained close to zero.

These results show that model ranking depends on the evaluation objective. The constant predictor performs best when every image contributes equally to the overall MAE. The balanced model performs better when reduced-visibility and clear-visibility conditions are given equal importance.

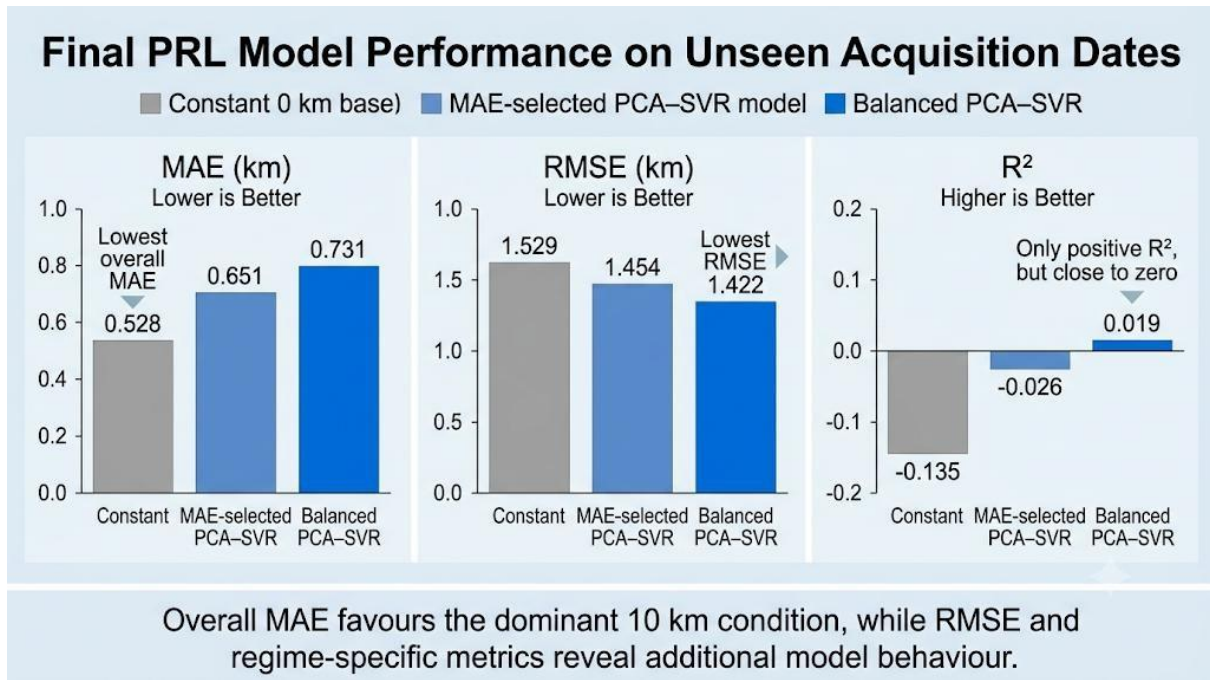


Figure 5.2: Overall MAE, RMSE and R² comparison for the three principal PRL models.

5.3 Reduced-Visibility Performance

The most scientifically important learned-model result was observed for the 22 images below 10 km. The constant baseline produced a low-visibility MAE of 3.045 km because it always predicted 10 km. The balanced PCA–SVR reduced this error to 2.491 km, an improvement of approximately 0.555 km.

The balanced-regime MAE gave equal importance to observations below 10 km and those labelled exactly 10 km. Under this measure, error decreased from 1.523 km for the constant baseline to 1.426 km for the balanced model, an improvement of approximately 0.096 km.

Table 5.3: Paired comparison of the balanced PCA–SVR with the constant baseline

Metric	Constant baseline	Balanced PCA–SVR	Difference	95% paired date-cluster interval	Interpretation
Overall MAE	0.528	0.731	+0.203	0.066 to 0.313	Baseline better
Low-visibility MAE	3.045	2.491	−0.555	−0.683 to −0.398	Balanced model better
Balanced-regime MAE	1.523	1.426	−0.096	−0.170 to −0.012	Balanced model better

The low-visibility and balanced-regime intervals excluded zero. This indicates that the observed improvements were supported across the available acquisition dates rather than being caused by only one convenient split. The improvements remain modest and should be interpreted within the limitations of the small dataset.

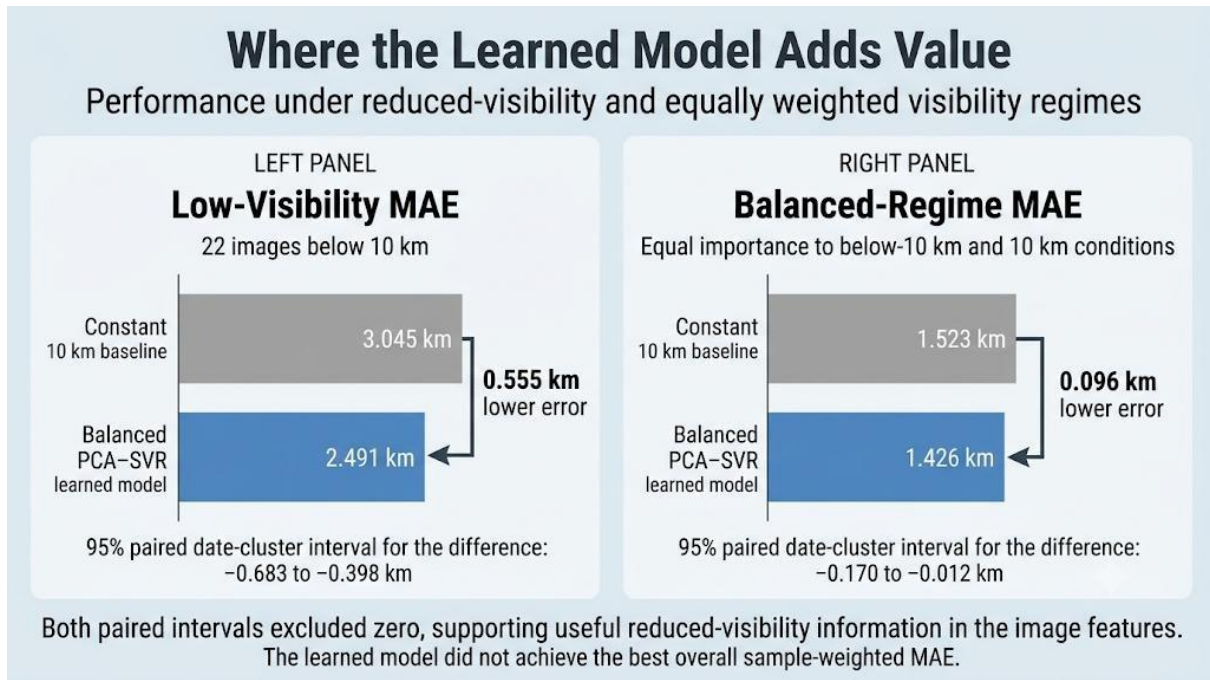


Figure 5.3: Reduced-visibility and balanced-regime error comparison.

5.4 Effect of Dataset Imbalance

The difference between overall and reduced-visibility results is explained by the PRL target distribution. Of the 127 observations, 105 were labelled exactly 10 km. A constant prediction of 10 km therefore produced no error for 82.7% of the dataset.

The learned model lowered some predictions when visual features indicated reduced clarity. This improved predictions below 10 km but introduced small errors on a large number of clear observations. Since overall MAE weights every image equally, these small errors accumulated across the 105 dominant-label images.

The constant baseline should therefore be regarded as the correct winner for the observed overall MAE. At the same time, its good result does not demonstrate sensitivity to atmospheric variation; it is produced by repeating the dominant label. The learned model provided additional information by responding to reduced-visibility images. The single 4 km image and ten 5 km images were insufficient for strong calibration at the lower end of the visibility range. Most held-out predictions remained near 10 km, with downward adjustments for some 5 km and 9 km cases.

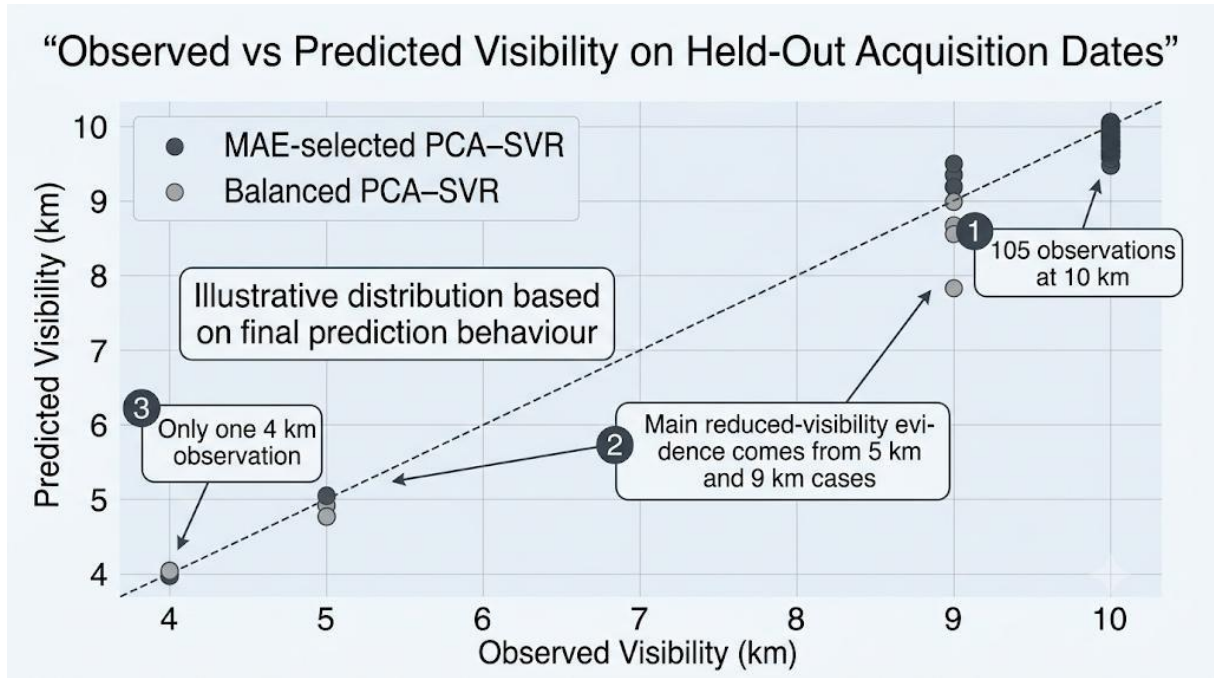


Figure 5.4: Observed and predicted visibility for held-out PRL images.

5.5 Scientific Interpretation

The results should not be reduced to either “the model outperformed the baseline” or “the model failed.” Both interpretations are incomplete. The constant 10 km baseline achieved the lowest overall MAE because the dataset was dominated by exact 10 km observations. The balanced PCA-SVR accepted small errors on some of these clear cases in exchange for lower error below 10 km. The date-cluster analysis supports the conclusion that the image representation contained useful reduced-visibility information.

Image-only PCA-SVR was selected as the final target-domain model. Weather fusion did not add stable improvement on unseen PRL dates. Possible reasons include the limited range of local weather conditions, repeated weather measurements across nearby images and the ability of the image itself to represent both atmospheric and illumination changes.

The transfer-weighted Ridge experiment also underperformed the local models. This indicates that simple transfer of source visibility and weather relationships was insufficient to overcome differences between the online and PRL domains. DINOv2 still provided representation-level transfer because it was pretrained on broad visual data.

The most defensible conclusion is:

The learned image model did not outperform the constant 10 km reference on overall sample-weighted MAE, but it demonstrated statistically supported sensitivity to reduced-visibility conditions.

The completed system should therefore be presented as a fixed-scene research prototype and reproducible baseline, not as a universal or safety-critical visibility instrument.

CHAPTER 6

SOFTWARE IMPLEMENTATION AND REPRODUCIBILITY

6.1 Software Architecture

The project was implemented as a modular Python workflow covering data verification, feature extraction, dataset preparation, model development, final evaluation and single-image prediction.

The processing flow is:

Raw images and metadata → **Verification** → **DINOv2 embeddings** → **Prepared arrays** → **Model development** → **Nested evaluation** → **Final results and deployment bundles**

Raw images, large embeddings, processed arrays and trained binary models were maintained locally. Source code, aggregate results, documentation and figures were organised separately so that the methodology remained reproducible without unnecessarily increasing repository size.

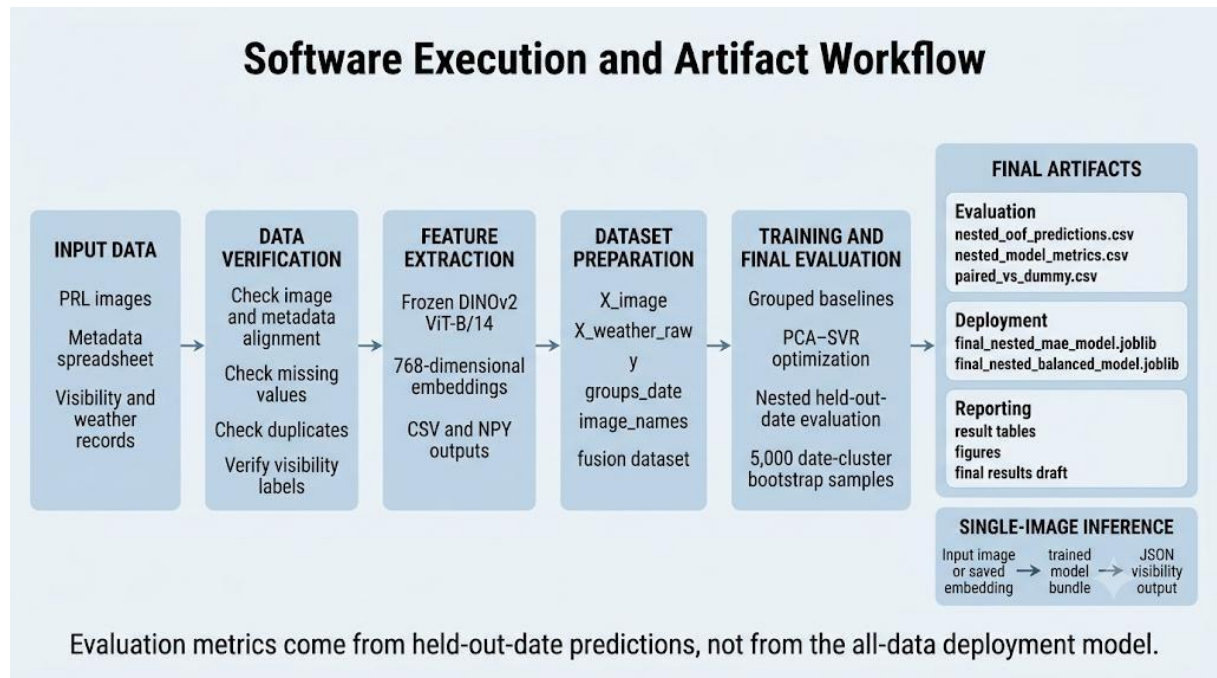


Figure 6.1: Software execution and artifact-generation workflow.

6.2 Core Scripts

Table 6.1: Main software components

Script	Responsibility
run_prl_pipeline.py	Runs verification, preparation, grouped development training, evaluation and result generation

scripts/run_final_lockin.ps1	Executes tests, final nested benchmark, bootstrap analysis and final reporting
01_verify_prl_dataset.py	Checks images, metadata, embeddings, duplicates, missing values and label distribution
02_extract_prl_features.py	Extracts DINOv2 ViT-B/14 embeddings
03_prepare_prl_dataset.py	Aligns metadata and features and writes model-ready arrays
04_finetune_fusion.py	Runs grouped baseline, Ridge, XGBoost, fusion and transfer experiments
06_predict.py	Performs single-image or saved-embedding visibility inference
08_optimize_accuracy.py	Runs the broad PCA-SVR development search
09_nested_grouped_benchmark.py	Runs the final nested grouped evaluation
10_evaluate_nested_results.py	Calculates confidence intervals and paired comparisons
11_generate_final_paper_results.py	Produces locked result tables, figures and report-ready text
final_evaluation.py	Stores the frozen candidate grid, metrics and evaluation utilities

6.3 Training and Evaluation Workflow

The complete local workflow is executed in the following order:

1. Verify that images, metadata and embeddings are complete and aligned.
2. Extract or refresh DINOv2 embeddings when required.
3. Prepare the final NumPy arrays, fusion dataset and acquisition-date groups.
4. Run grouped development experiments.
5. Run the frozen nested held-out-date benchmark.
6. Generate 5,000 date-cluster bootstrap samples.
7. Produce final result tables, plots and model bundles.

The authoritative evaluation outputs are generated from nested held-out-date predictions. Model bundles fitted later using all 127 observations are intended for inference only.

Table 6.2: Principal final artifacts

Artifact	Purpose
nested_oof_predictions.csv	Final per-image held-out-date predictions
nested_model_metrics.csv	Final point estimates for evaluated models
outer_fold_selections.csv	Candidate selected in each outer fold
metric_confidence_intervals.csv	Date-cluster confidence intervals
paired_vs_dummy.csv	Paired comparisons with the constant baseline
per_date_metrics.csv	Performance calculated separately by acquisition date
final_nested_mae_model.joblib	All-data model selected using the MAE objective
final_nested_balanced_model.joblib	All-data model selected using the balanced objective

6.4 Single-Image Prediction

The prediction script supports two input routes:

- direct extraction of DINOv2 features from a new image; or
- retrieval of an existing embedding from the saved feature file.

The script loads the selected model bundle, applies the stored scaler and PCA transformation and returns a visibility estimate. The output also records the image name, model identifier, embedding source and a research-use warning.

An example output structure is:

```
{
  "image_name": "PRL_0001.jpg",
  "predicted_visibility_km": 9.6348,
  "model_name": "image_svr_pc2_C30.0_e0.75",
  "model_type": "optimized_image_svr",
  "embedding_source": "image_extraction_torchhub_dinov2_vitb14",
  "warning": "Research prototype; do not use for safety-critical decisions."
}
```

Weather inputs may remain available in the command-line interface for compatibility with earlier fusion models. They do not alter the result when an image-only model bundle is loaded.

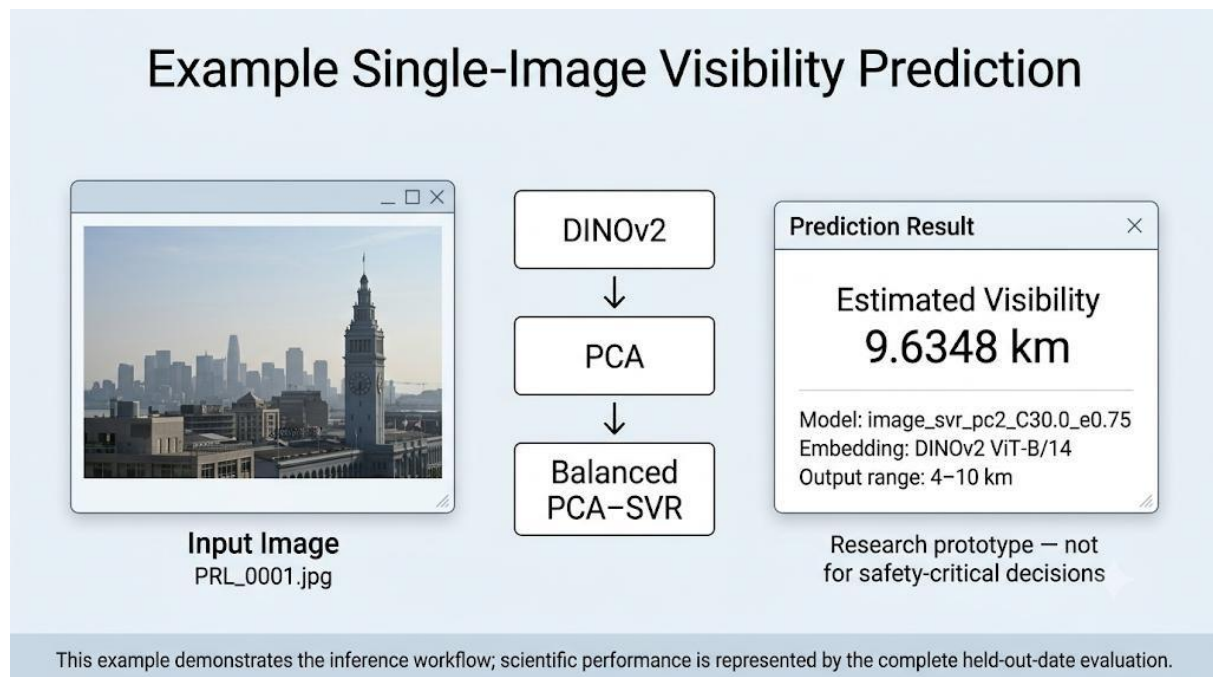


Figure 6.2: Example single-image visibility prediction output.

6.5 Models, Results and Tests

Table 6.3: Automated verification tests

Test	Status
Metadata and embeddings have matching row counts	Passed
Out-of-fold predictions cover every sample	Passed
Prepared dataset arrays have valid shapes	Passed
Selected model bundle returns a finite prediction	Passed
Engineered meteorological features are finite	Passed
Final metrics include regime and date measures	Passed

Frozen candidate grid is small and unique	Passed
---	--------

Version-control milestones preserved the baseline workflow, PCA–SVR optimization and final nested lock-in. The repository also separates exploratory outputs from final authoritative results. The implementation is therefore reproducible at the code and aggregate-result level, provided that the local images, embeddings and trained model files are restored in the documented directories.

CHAPTER 7

LIMITATIONS AND FUTURE WORK

7.1 Current Limitations

Table 7.1: Current limitations and implications

Limitation	Implication
127 images but only 23 acquisition dates	Effective independent sample size is small
105 images labelled exactly 10 km	Overall MAE strongly favours the constant baseline
Only 22 images below 10 km	Reduced-visibility calibration is limited
Only one 4 km observation	Severe low-visibility behaviour cannot be estimated reliably
Single camera and fixed scene	External generalisation is unknown
Collection over approximately one month	Seasonal variability is not represented
Afternoon observations only	Morning, evening and night-time conditions are absent
No calibrated visibility-instrument comparison	Predictions are not instrument-validated
Source and target domains differ	Simple source transfer did not improve local performance

The model may also be affected by camera exposure, sunlight, cloud cover, lens condition and changes in the observed scene. The system should not be used for aviation, road safety, emergency response or other safety-critical decisions.

7.2 Future Data Collection

The highest priority is to collect more independent atmospheric dates, especially observations below 10 km. Additional images from the same date are less valuable than observations from new haze, fog or reduced-contrast events.

Future collection should:

- cover multiple seasons;
- include morning and evening observations;
- record more examples between 4 km and 9 km;
- document rainfall, cloud state and lens contamination;
- include wind, pressure and additional atmospheric variables where available; and
- preserve the fixed-camera protocol for direct comparison.

Adding a second viewpoint or observation location would allow the system to be evaluated beyond one fixed scene.

7.3 Model and System Improvements

With a larger local dataset, image–weather fusion should be reassessed. The current negative fusion result may be related to narrow weather ranges, repeated measurements and limited independent dates rather than a general lack of meteorological value.

Other future improvements include:

- horizon-focused or landmark-focused image regions;
- scene-quality checks for camera movement and lens obstruction;
- prediction intervals for individual estimates;
- validation on a second location;
- testing recent compact vision encoders;
- automated acquisition and real-time processing; and
- a dashboard for time-series visibility monitoring.

Further parameter tuning on the existing 127 images should not be the immediate priority. New independent data and external validation are likely to provide greater scientific value.

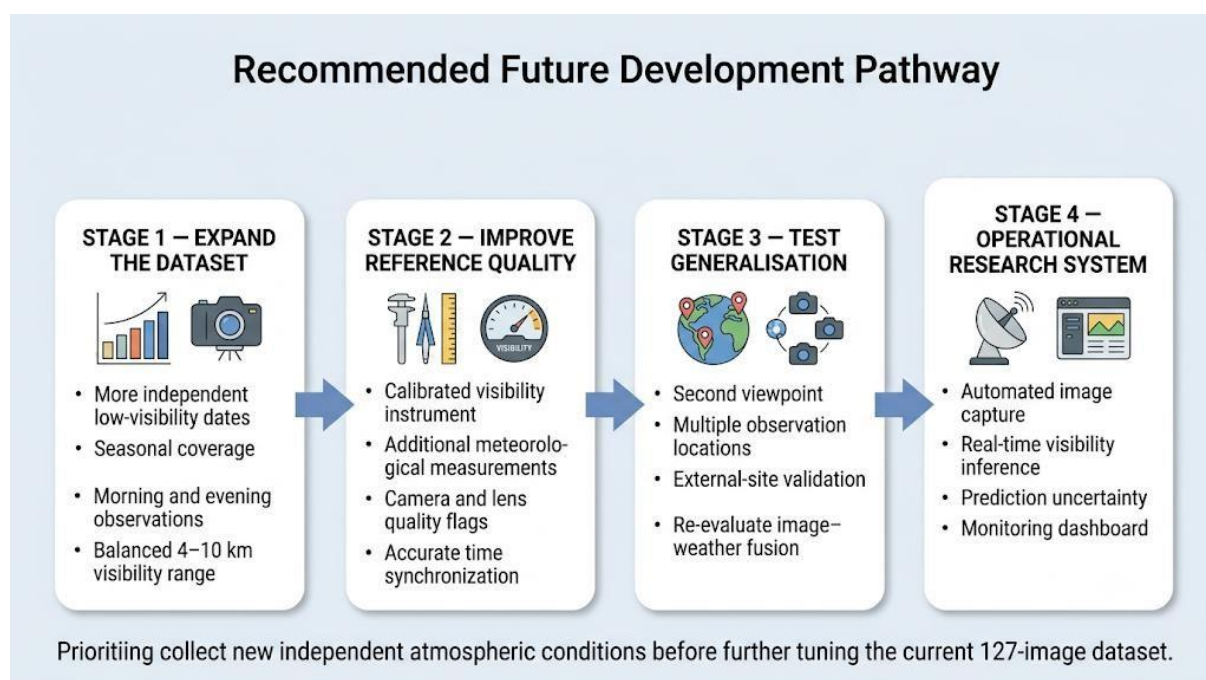


Figure 7.1: Recommended future development pathway for the visibility-monitoring system.

7.4 Future Atmospheric Applications

Aerosol Optical Depth may be explored in future work, but it requires an independent reference such as a quality-controlled sun photometer, satellite aerosol product or other validated AOD source.

Horizontal visibility and vertical column AOD are not equivalent measurements. A research-grade AOD study would require:

- temporal and spatial matching with independent AOD observations;
- wavelength harmonisation;
- cloud and fog screening;
- humidity treatment;
- information about aerosol vertical distribution; and
- grouped evaluation on unseen dates or sites.

Any visibility-derived AOD calculation without such reference data should be described only as a proxy or sensitivity estimate, not as measured or validated AOD.

CHAPTER 8

CONCLUSION

8.1 Summary of Work

This project developed a complete research workflow for estimating atmospheric visibility from ground-based imagery at the Physical Research Laboratory, Ahmedabad.

A 14,906-image SkyFinder source subset was prepared to evaluate visibility learning using image and meteorological information. A local dataset of 127 fixed-view GoPro images was collected across 23 acquisition dates, together with visibility labels and supporting atmospheric observations.

Frozen DINOv2 ViT-B/14 features were extracted from the images. PCA was used to reduce the 768-dimensional representation, and Ridge, XGBoost and SVR models were evaluated using weather-only, image-only and fusion configurations.

The final local model family was image-only PCA–SVR. Performance was evaluated by holding out complete acquisition dates, with preprocessing and model selection performed only on the remaining dates. Date-level bootstrap analysis was used to quantify uncertainty.

8.2 Main Findings

The large source dataset demonstrated that outdoor images and meteorological information contained useful visibility-related patterns. The image–weather fusion model produced the best source-domain result, with an MAE of 1.550 km and R^2 of 0.604.

On the local PRL dataset, the constant 10 km baseline achieved the lowest overall MAE of 0.528 km. The MAE-selected PCA–SVR achieved 0.651 km, while the balanced PCA–SVR achieved the lowest RMSE of 1.422 km and an R^2 of 0.019.

The overall result was strongly influenced by the label distribution, since 105 of the 127 observations were exactly 10 km. The balanced PCA–SVR nevertheless reduced low-visibility MAE from 3.045 km to 2.491 km and improved balanced-regime MAE from 1.523 km to 1.426 km. Both improvements were supported by paired date-cluster confidence intervals.

Weather fusion and simple source-weighted transfer did not improve final PRL generalisation. The most consistent local representation consisted of a small number of PCA components derived from DINOv2 image features.

8.3 Final Outcome

The project establishes a reproducible fixed-scene baseline for atmospheric visibility estimation from ground-based imagery.

The learned model did not outperform the constant reference on the primary sample-weighted MAE. However, it demonstrated useful and statistically supported sensitivity to reduced-

visibility conditions. The correct interpretation is therefore neither that the model universally outperformed the baseline nor that the image information was ineffective.

The completed work provides:

- a verified local dataset;
- a documented source-to-target development process;
- a leakage-controlled evaluation methodology;
- final held-out-date predictions and uncertainty analysis;
- deployable visibility-model bundles; and
- a repeatable single-image inference workflow.

Future progress should prioritise additional independent low-visibility dates, broader seasonal coverage, multiple observation locations and comparison with calibrated reference instruments. With these extensions, the current pipeline can serve as the foundation for a more general and operational camera-based atmospheric monitoring system at PRL.